



AI ORIENTATION

AND PROMPT ENGINEERING



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ARTIFICIAL INTELLIGENCE

- Artificial Intelligence (AI) is the umbrella term for systems that can perform tasks traditionally requiring human intelligence
 - Decision making, language understanding, and pattern recognition
- AI is a broad field in computer science
- Includes rule-based and learning-based systems

MACHINE LEARNING

- Machine learning(ML) is a subset of AI
- ML allows systems to learn from examples instead of being explicitly programmed
- The more data the system sees, the better it typically performs
- Performance improves over time

GENERATIVE AI

- Generative AI (GenAI) is a subset of machine learning
- GenAI generates new outputs that resemble human-created content
- Capable of producing text, images, code, audio, video, etc.
- Based on large language and foundation models

PROMPT ENGINEERING

- Prompt engineering is designing effective inputs to AI systems to get the desired or intended output
- It is the practice of crafting prompts that clearly communicate what you want the model to do
- Prompts guide model behavior and output
- Acts as the user interface of GenAI
- Combines instructions, context, and examples
- Critical for AI effectiveness

ZERO-SHOT PROMPTING

- Zero-shot prompting asks the model to perform a task without giving examples and relying entirely on what it learned during training
 - "Summarize this article in three sentences"
- This is a fast and flexible prompt style that works well for common tasks that likely don't need an example to get the intended result
- The model relies completely on prior training
 - If a model has not been trained on the details of the request, the result will likely not be what was intended

CONSTRAINTS

- Constraints help narrow the model's response space, leading to more predictable and useful outputs
- It lets us limit format, tone, length, etc.
- It improves consistency and reduces ambiguity
- Adding constraints is essential for effective prompt engineering
 - "Summarize this story with a more joyful, upbeat tone in 6 sentences"

FINE TUNING VS CONDITIONING

Fine Tuning

- Model retraining with new data
- Long-term behavior changes
- Alters infrastructure
- Permanently alters the model's behavior

Conditioning

- Temporary guidance via prompts
- Context-based
- No model changes
- Shapes behavior only for the current interaction

INTERACTIONS & DIALOGUE STATE

- AI models track conversation context, which influences future responses
- Poor context management can lead to confusion or conversation drift
- The model remembers what you say, so use that wisely!
- Context accumulates over time and impacts accuracy and relevance
- Conversation history matters and must be managed intentionally

INSTRUCTIONS AND GUIDELINES

- When building a prompt, you want to give explicit instructions to improve the results
- Clear instructions act like guardrails, guiding the model toward acceptable and useful responses
- Ambiguous instructions create ambiguous answers
 - "Create a meal planning app with a calendar"
 - "Create an application where users can add and organize meal recipes onto a repeating calendar"
- Better instructions prevents undesired results

HALLUCINATIONS

- AI Hallucinations occur when a model generates plausible-sounding but false information
- These are confident but incorrect outputs
- The result of pattern-based generation, not intentional deception
- We see lots of hallucinations with code! That is why vibe coding is an important skill

RESPONSIBLE USAGE

- AI should assist, not replace human judgement
- Responsibility still remains with the user
- AI accelerates decision, but the humans own them
- Human oversight is required to verify outputs and avoid misuse and bias
- Respect ethical boundaries

SECURITY

- AI systems can expose sensitive information if misused
- Always be aware of security risks and do not input sensitive data
- Protect proprietary information
- Follow organizational policies
- Beware of prompt injection
 - Treat AI like a public system, not a private one

SUMMARY

- AI is the umbrella concept
- ML enables learning from data
- GenAI creates new content
- Prompting shapes outcomes
- AI has limitations and risks
- Responsible and secure use is essential